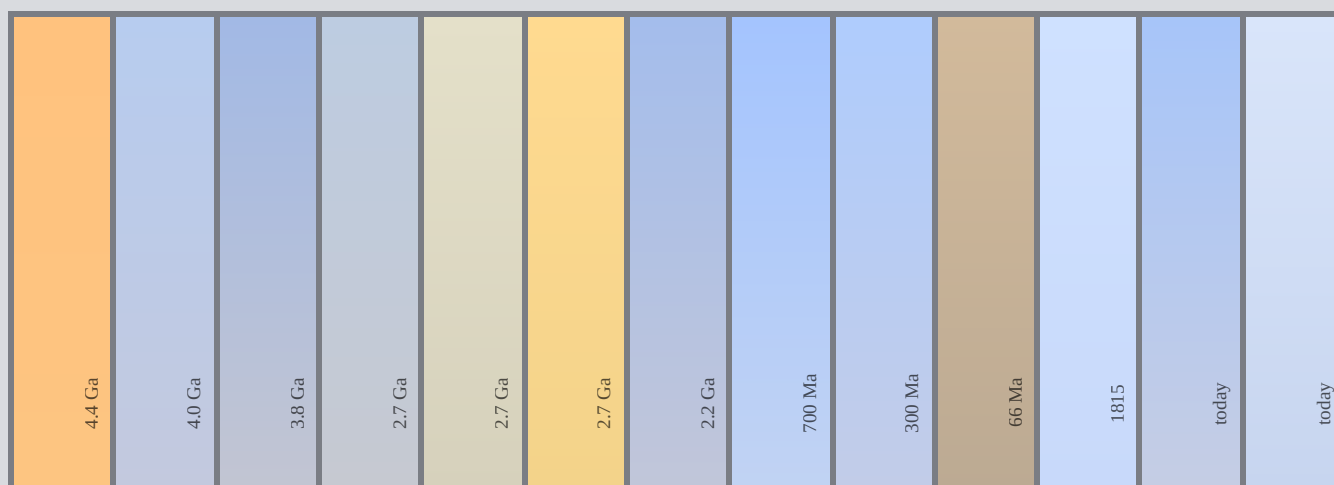


The colour of Earth's sky, *4.4 billion years to today*

A first-pass spectral radiative-transfer reconstruction of what the sky looked like from the ground at thirteen moments in Earth's history: straight up and at the horizon, at the equator, mid-latitudes and the poles, at noon, sunset and dusk.



Mid-latitude noon sky, zenith (top) to horizon (bottom), oldest at left.

Brightness is shown relative to today's clean sky. Tap a column to jump to that epoch.

Has this been done before?

Not as such. A search of the Valency corpus (arXiv, PubMed, EarthArXiv and others) finds the two halves of the problem solved separately but never joined. On one side, atmospheric-optics work computes modern sky colour rigorously: Lynch & Mazuk (2005) derive CIE chromaticities of the sky, Sun and distant objects from MODTRAN spectra; Peyvandi et al. (2016) simulate 5.6 million daylight spectra with SBDART to map how aerosol and ozone shift sky colour; Lee & Mollner (2017) show with Monte Carlo simulation that vivid solar-horizon twilight colours need aerosol while the purest antisolar colours occur in a purely molecular atmosphere. On the other side, paleoatmosphere work constrains composition: Catling & Zahnle

(2020) for the Archean (O_2 below 10^{-6} of present, CO_2 10–2,500× modern, CH_4 10^2 – 10^4 × modern), and Arney et al. (2016), Mak et al. (2023) and Goldblatt et al. (2024) for the organic-haze episodes. Arney's "Pale Orange Dot" computes what a hazy Archean Earth looks like *from space*, as an exoplanet spectrum, and that is as close as the literature comes. Nobody has computed the view from the ground, or the horizon-versus-zenith, latitude, and sunset structure. So this report does.

How the colours were computed

For each epoch the atmosphere is built from components with their own vertical profile and spectral optical depth: Rayleigh-scattering gas (N_2 , O_2 , CO_2 , CH_4 , H_2O in their epoch-appropriate partial pressures), ozone (Chappuis band), tropospheric aerosol, a stratospheric sulfate layer, a tholin-like fractal organic haze (strongly blue-absorbing, forward-scattering), and impact soot. Sky radiance is integrated numerically along each line of sight through a spherical-shell atmosphere, so grazing sunlight at twilight is handled properly, with multiple scattering and surface reflection added from a delta-Eddington two-stream solution. The young Sun is made fainter and slightly cooler following standard solar evolution (70% luminosity at 4.4 Ga). Spectra from 380–780 nm are converted to CIE XYZ and shown as sRGB. The model reproduces measured modern values (clear zenith 10,000–20,000 K, horizon 6,000–8,500 K, setting Sun ~1,700 K, pink antisolar Belt of Venus) before being pointed backward.

Findings in brief

Epoch	Noon zenith, mid-lat	Horizon	Sunset
4.4 Ga, 30-bar CO_2	Peach-white, 3,300 K, shadowless	Same as zenith	Whole dome dims to orange; Sun rarely visible
4.0 Ga, 0.5-bar CO_2	Pale milky blue, 10,500 K	Milk white	Deep red Sun; cream twilight (no ozone)

3.8 Ga, clear Archean	Blue, 13,000 K	White	Modern-like, then yellow-beige twilight zenith
2.7 Ga, thick haze	Cream-tan, 5,600 K	Cream-tan	Vermilion horizon, amber sky, long orange dusk
2.2 Ga, post-GOE	Blue, 13,700 K	White	Blue twilight zenith returns
700 Ma, Snowball	Deep blue, 17,000 K	Blue-white, stays blue	Golden, pale
300 Ma, 33% O ₂	Blue, 13,600 K	White, slightly brighter	Modern-like; smoky more often
66 Ma, impact winter	Dim amber-beige, 4,600 K	Beige	None; dome fades to brown
1815, volcanic year	Milky blue-white, 9,300 K	Slightly bluer than zenith	Salmon Sun, pink-lavender afterglow
Today, clean	Blue, 15,000 K	White, 8,400 K	Orange horizon, pink antisolar

Horizon versus zenith

In a clean molecular atmosphere the zenith is blue and the horizon white because the horizon path is long enough that blue is scattered back out before reaching the eye and multiple scattering fills in the other colours. The size of that gradient is the clearest single diagnostic of atmospheric state. It is largest in the cleanest atmospheres (Snowball Earth, today's clean air: 8,000–13,000 K of colour-temperature difference), shrinks as Rayleigh depth rises (late Hadean), collapses to nothing when a broadband absorber or thick scatterer sits overhead (hazy Archean, impact winter, 30-bar Hadean), and actually *reverses* under a stratospheric sulfate veil or heavy pollution, where the zenith becomes whiter than the horizon.

Equator, mid-latitudes, poles

Latitude enters through solar elevation and surface albedo. A high Sun lights the sky from above through a short path, giving a brighter but paler zenith

(equatorial noon $\sim 9,000$ K); a low Sun sends light through more air, so the polar summer zenith is a deeper, darker blue ($\sim 19,000$ K today, $20,000$ K over Snowball ice) while the polar horizon is warmer and yellower. The same geometry makes the poles the place where any overhead absorber is most visible: the hazy Archean sky is cream at the equator but distinctly orange-yellow at the poles, and the impact-winter Sun that is still a pale disc at the equator is gone at 75° latitude. Snow and ice brighten and blue-shift the whole dome by reflecting light back into the atmosphere, which is why polar and Snowball horizons stay blue where a mid-latitude land horizon goes white.

Particulates

Three particle regimes appear in the record and they look different from one another. Non-absorbing sulfate high in the stratosphere (volcanic years) whitens the day sky and produces long, luminous pink and lavender twilights, because the layer is still sunlit after the ground is dark. Absorbing organic haze (Neoarchean) or soot (impact winter) removes blue outright, turning the dome tan or amber with almost no directional structure, and dims the surface by factors of 3–20. Boundary-layer pollution or dust brightens and greys the zenith, and buries the setting Sun in murk before it reaches the horizon. Tropospheric aerosol is also what makes sunset colour vivid: the model confirms Lee & Mollner's result that a purely molecular sky yields a pale, yellow-orange horizon while any turbidity deepens it to red.

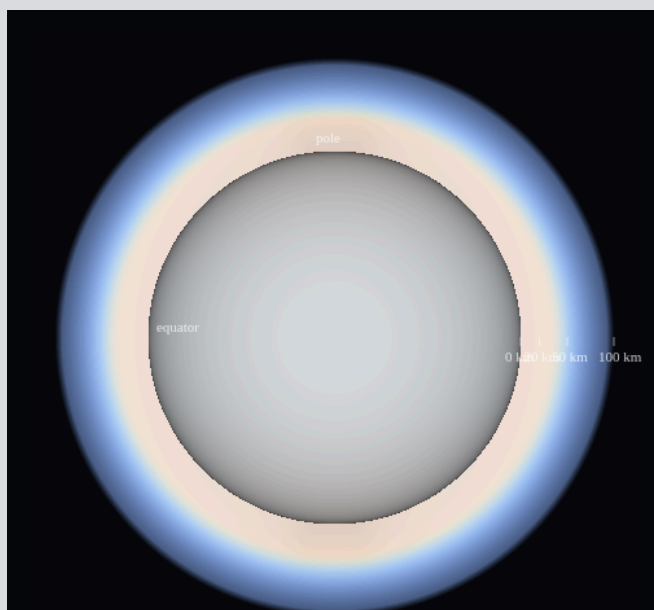
Sunsets and sunrises through time

The biggest surprise is ozone. Today the sky above a setting Sun grades from orange through pale yellow to a zenith that stays blue well into twilight; that persistent blue is not Rayleigh scattering but ozone absorption of yellow and orange light along the grazing path. Before the Great Oxidation Event there was no ozone, so every sunset for the first two billion years ended with the zenith fading through cream and beige to orange, and the sky never turned that deep twilight blue. The return of a blue dusk is, in this model, a direct

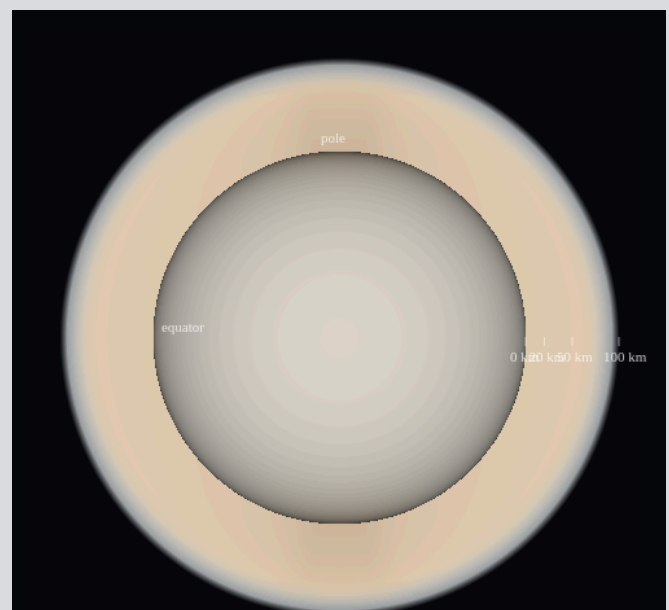
optical consequence of oxygenation at ~ 2.4 – 2.2 Ga. Beyond that: Hadean sunsets were not events at the horizon but a slow dimming of the whole glowing dome; hazy Archean sunsets were the longest and most saturated in Earth's history, vermillion at the horizon and amber overhead; Snowball sunsets were brief and golden; the sunsets of the impact winter did not exist as colour at all; and the reddest, most theatrical skies of the Phanerozoic belong to volcanic years like 1815 and 1883.

The atmosphere from space

To show latitude and altitude at once, these renderings look at Earth from space with the Sun behind the viewer at the equinox, so the solar zenith angle at every point equals its latitude (equator at the centre line, poles at top and bottom). The shell around the disk is the atmosphere's limb: the colour of light scattered toward the viewer along a tangent ray at each altitude, computed with the same model. The atmosphere is drawn 30 times too thick, so 100 km spans half an Earth radius; a real limb is a hairline. The disk shows the planet's reflected colour over a dark ocean (over ice for Snowball Earth), with limb brightness compressed so faint upper layers remain visible.



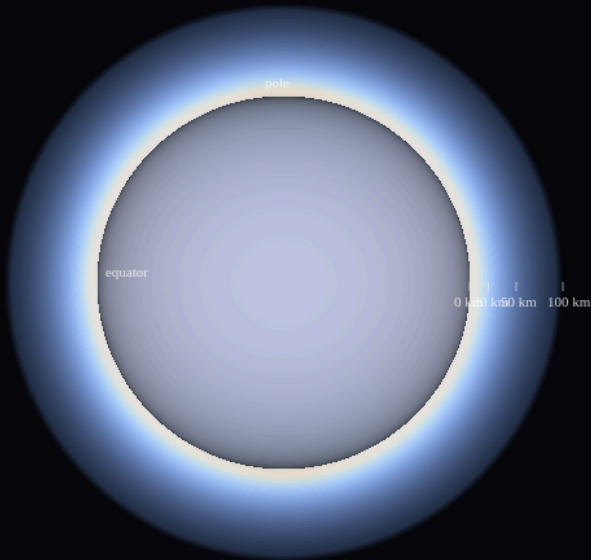
Early Hadean, ~ 4.4 Ga
30-bar CO_2 Hadean. The shell is an opaque, luminous white-peach almost all the way up, because tangent



Neoarchean, ~ 2.7 Ga (thick haze)
Hazy Neoarchean. The organic haze colours the entire limb tan: every tangent ray below ~ 60 km crosses the

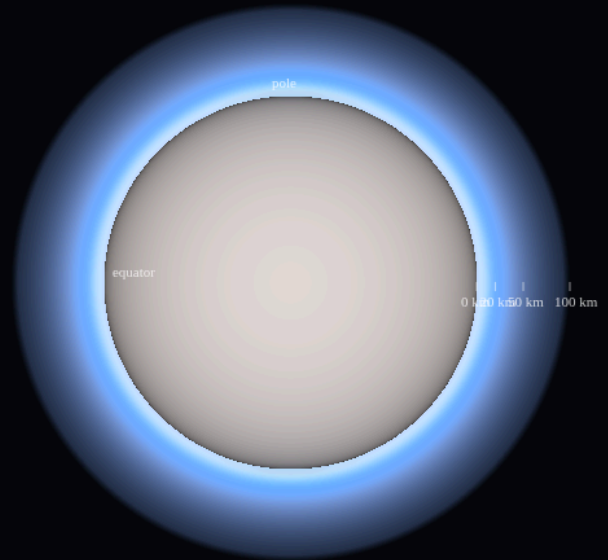
rays at any tangent height below ~60 km pass through many optical depths of CO₂; only the outermost fringe thins to blue. The disk is bright and nearly white: a cloudless, Rayleigh-only Venus.

layer (centred ~45 km) twice. Only above it does the shell turn a faint, dusty blue. The disk is the pale orange dot, though from space it is a subtle cream-orange rather than a vivid one.



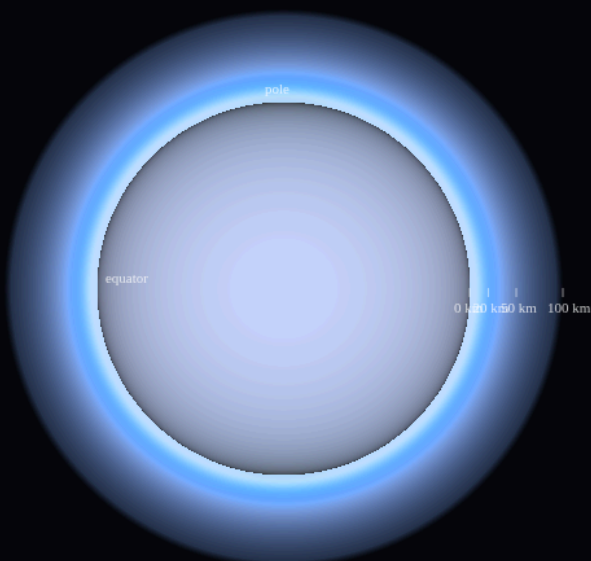
Early Archean, ~3.8 Ga

Clear early Archean. Structurally modern: white-blue near the surface, deep blue above, fading to black. With a little more CO₂ and no ozone the shell is marginally paler than today's and the disk very slightly whiter.

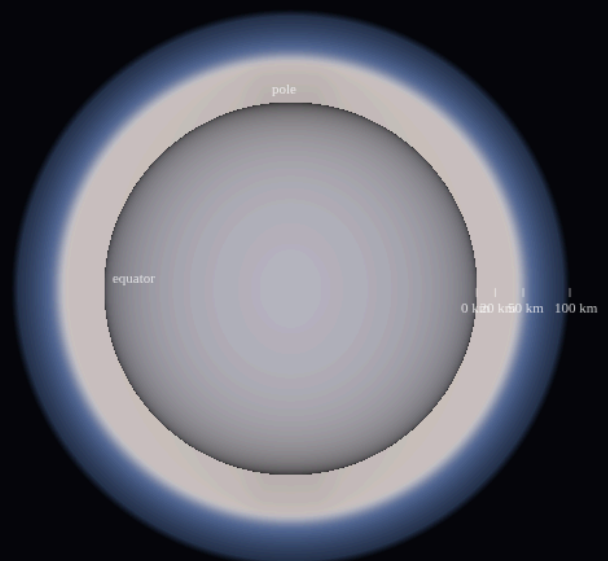


Snowball Earth, ~700 Ma

Snowball Earth. The bluest shell in the set above a blazing white disk of ice. The polar limb stays bright because the frozen surface reflects so much light back into the atmosphere.



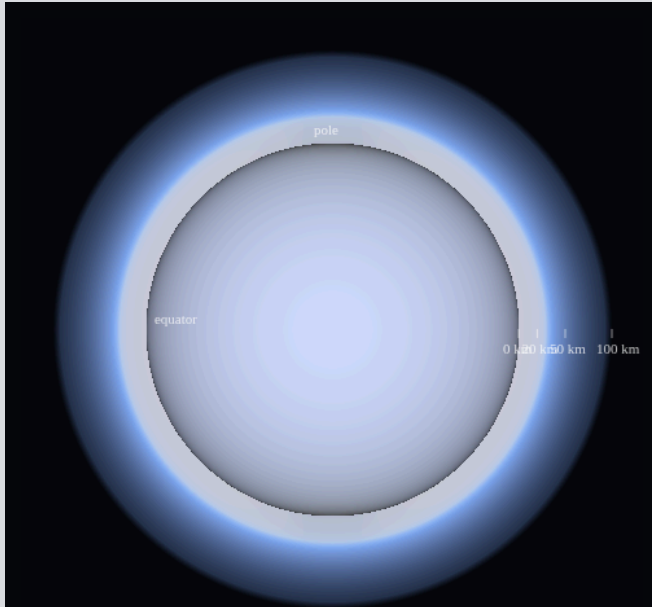
Carboniferous, ~300 Ma



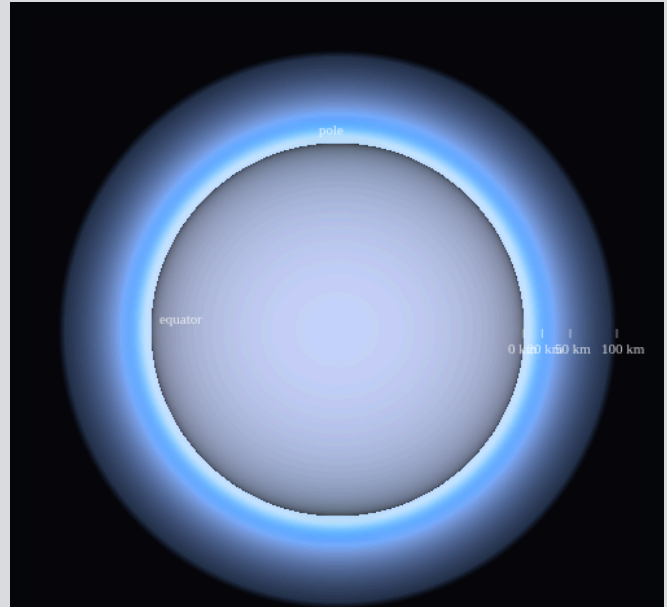
K-Pg impact winter, 66 Ma

Carboniferous, 33% O₂. The extra oxygen raises total pressure by about a tenth, so the limb is marginally thicker and brighter than today's, but to the eye it is indistinguishable from modern Earth: once ozone and clean air are in place, the sky from space has been the same pale blue dot for two billion years.

Impact winter. A dim brown-grey band from the soot and sulfate layer occupies the lower and middle shell; above ~40 km the untouched upper atmosphere is still faintly blue. The disk is a dull grey-lilac, and dark at high latitudes.



Volcanic year (Tambora 1815 / Toba-lite)
Volcanic year. A milky white sulfate band near 20 km sits between the white-blue troposphere and the deep blue upper shell. The disk is brighter and whiter than clean modern Earth.

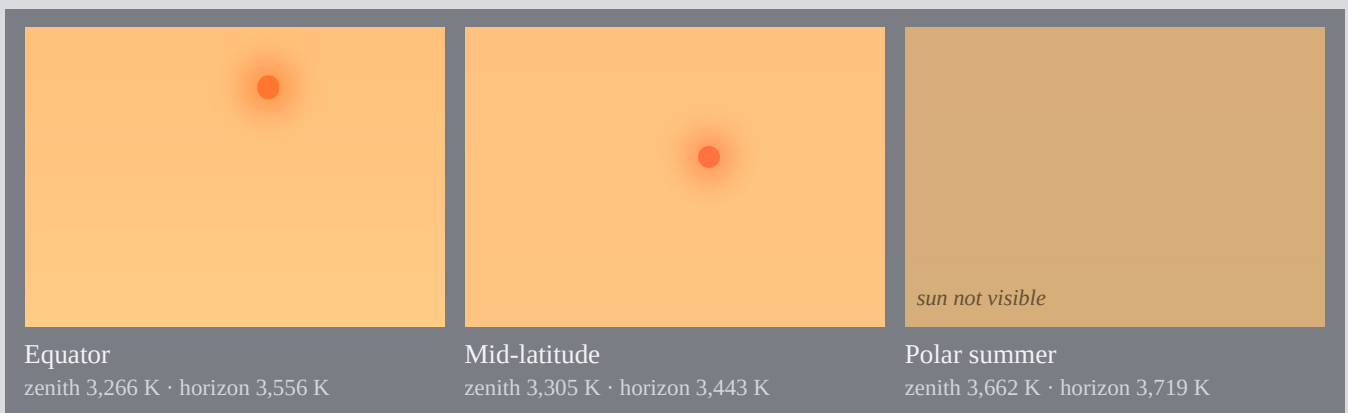


Modern, clean air
Modern Earth for reference: white-blue troposphere grading to a deep blue stratosphere and fading to black, brightest at the equator and darker toward the poles.

4.4 Ga

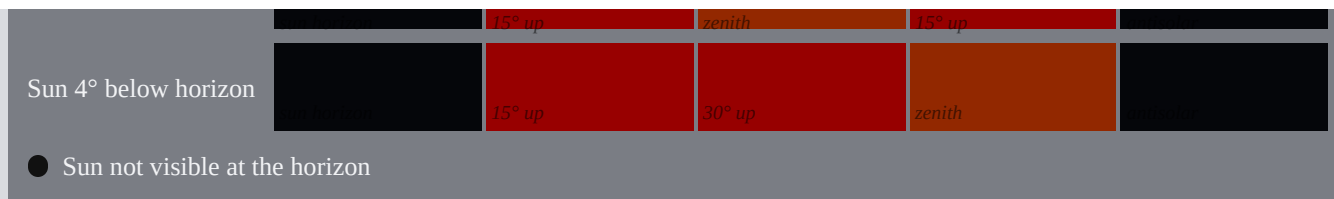
Early Hadean, ~4.4 Ga

post-magma-ocean CO₂/steam atmosphere (~30 bar CO₂)



Sun on horizon



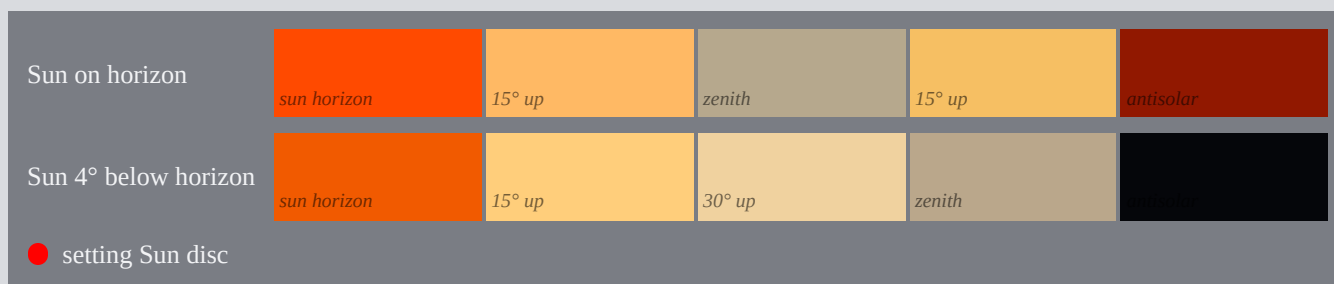


A ~30-bar CO₂ atmosphere has a Rayleigh optical depth near 5 at 550 nm and about 18 in the violet, so blue light is scattered back to space many times before it reaches the ground. What arrives is a diffuse, shadowless, peach-white glow of ~3,300 K, nearly identical from zenith to horizon, under a young Sun that is only 70% as bright. The Sun itself is a dim ember at the equator and vanishes entirely at mid and high latitudes: this is a Venus-like sky. Sunset is a brief, deep-orange dimming of the whole dome rather than an event at the horizon.

4.0 Ga

Late Hadean, ~4.0 Ga

CO₂ mostly drawn down into carbonate; ~1 bar N₂, ~0.5 bar CO₂



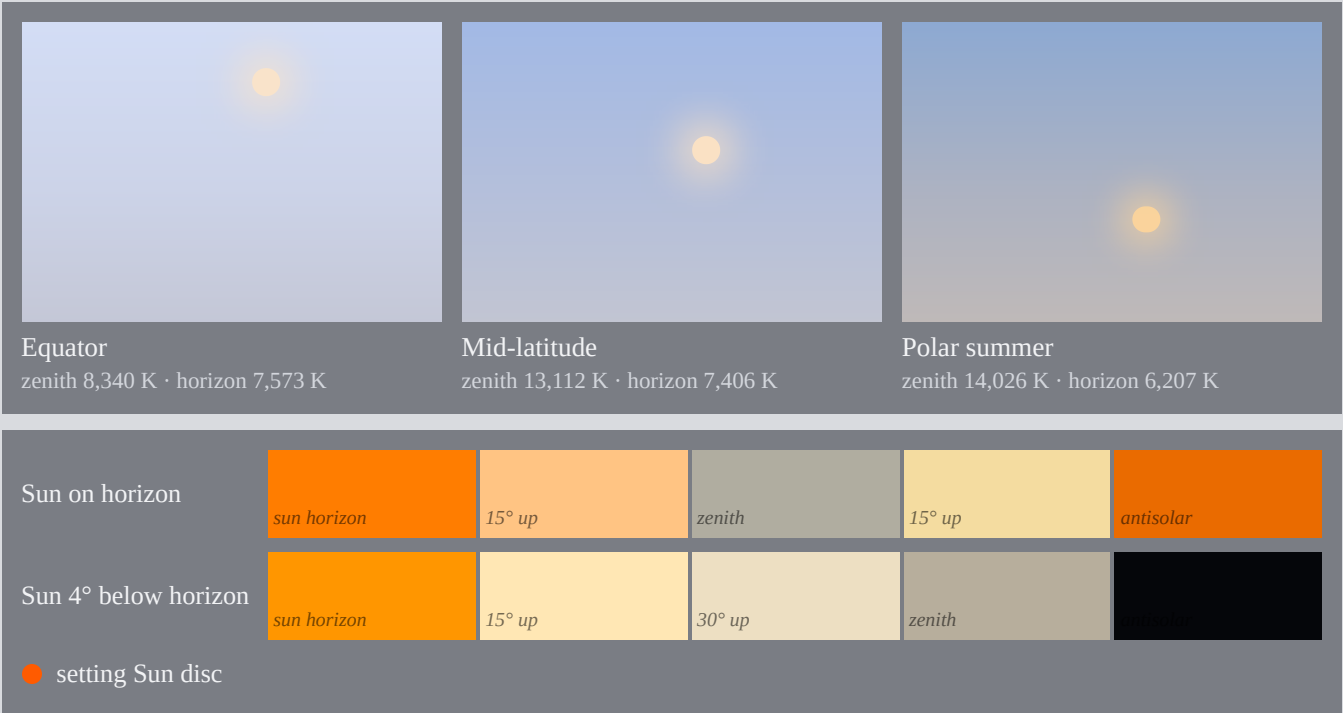
Once most of the CO₂ has gone into carbonate, Rayleigh scattering falls to roughly twice the modern value. The sky turns pale blue at the zenith but conspicuously whiter than today (7,000–10,500 K versus 9,000–15,000 K),

because the extra scattering fills the dome with multiply-scattered white light. Horizons are milky. The setting Sun is deep red and the twilight zenith, with no ozone yet, is cream rather than blue.

3.8 Ga

Early Archean, ~3.8 Ga

haze-free, anoxic; CO₂ ~0.1 bar, CH₄ ~1000 ppm, no ozone

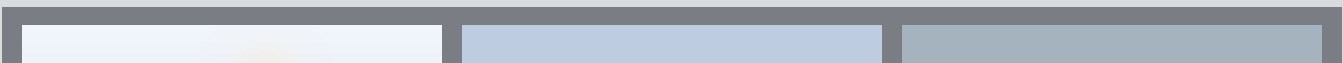


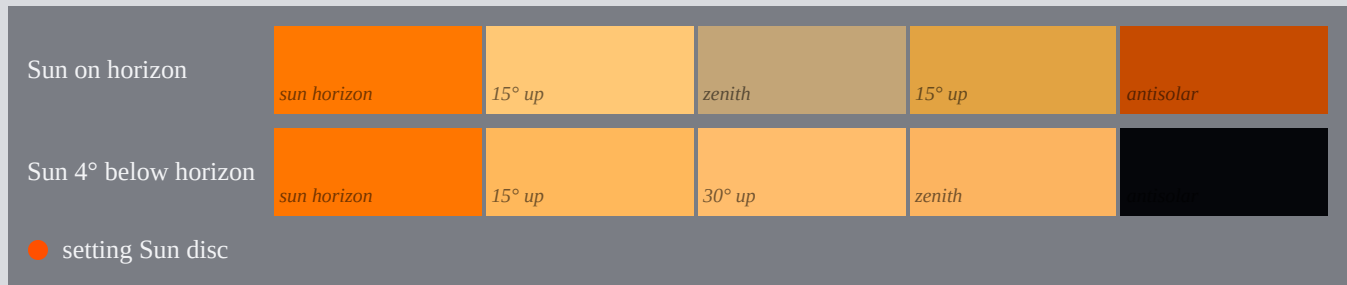
With CO₂ down to ~0.1 bar, noon skies are close to modern blue, only slightly whiter. The unmistakable difference is at dusk. The blue of the modern twilight zenith is produced by ozone absorbing yellow-orange light (the Chappuis band) along the long grazing path; with no free O₂ there is no ozone, so the Archean twilight zenith is a pale yellow-cream, and the whole sky fades from blue to beige to orange as the Sun sets. Sunsets themselves are much like today's, slightly redder from the extra CO₂ and dimmer Sun.

2.7 Ga

Neoproterozoic, ~2.7 Ga (thin haze)

biogenic CH₄/CO₂ ~0.1: intermittent thin organic haze



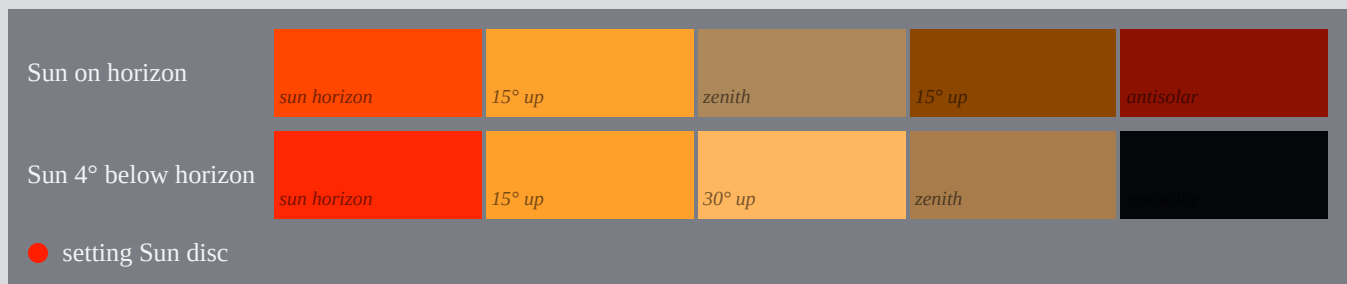


An intermittent organic haze at optical depth ~ 0.15 pushes the zenith toward a pastel, milk-blue (7,000–8,500 K) and lays a warm cast on the horizon; at the poles, where sunlight crosses the haze obliquely, the horizon turns visibly cream. Twilight is orange all the way to the zenith.

2.7 Ga

Neoproterozoic, ~ 2.7 Ga (thick haze)

"Pale Orange Dot": Titan-like fractal organic haze, $\tau(550\text{nm}) \sim 0.6$

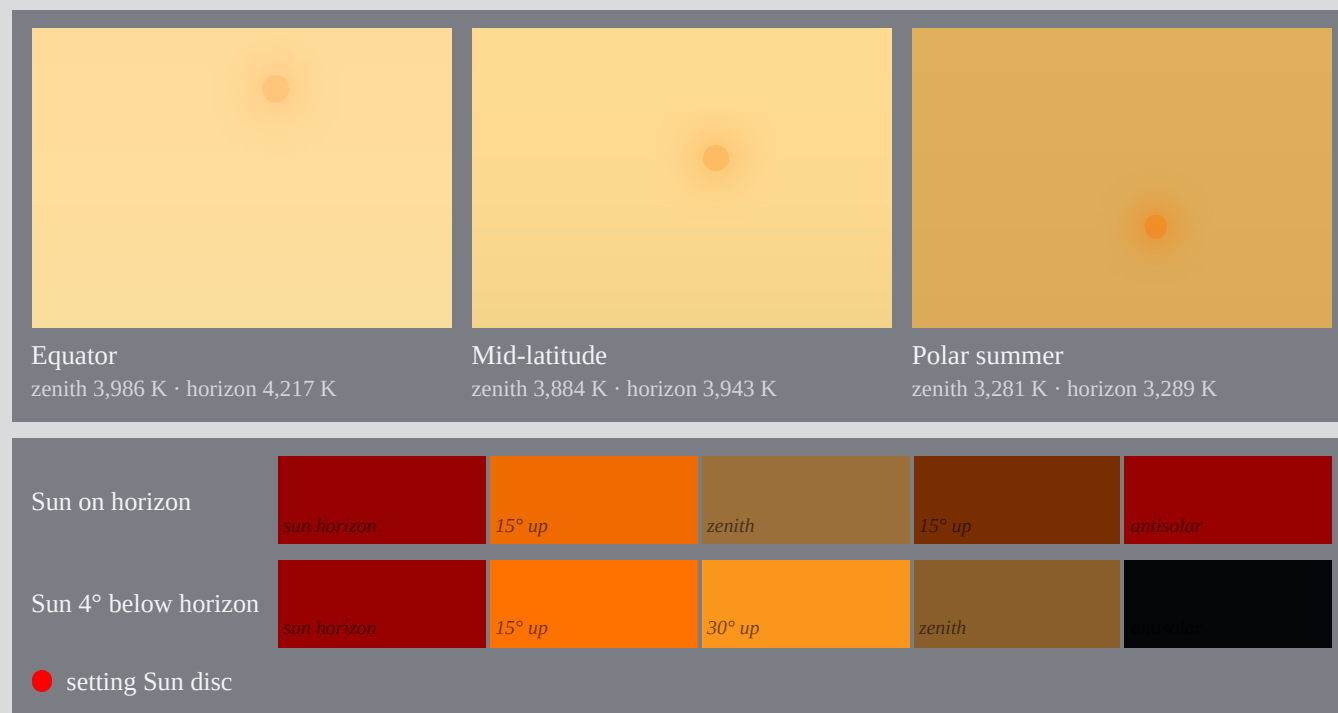


This is the 'Pale Orange Dot' seen from below. At optical depth ~ 0.6 the haze absorbs blue and scatters forward, so the entire dome is a warm cream to pale tan ($\sim 5,300$ K at the equator, $\sim 4,700$ K at the poles) with almost no zenith-to-horizon gradient: the sky looks like a bright overcast made of orange-tinted glass. The Sun is a soft, defined disc (haze scatters strongly forward) at $4,100$ K. Sunsets are deep and long: the solar horizon reaches vermillion, the sky 15° above the Sun is amber, and the antisolar sky is peach rather than pink.

2.7 Ga

Neoproterozoic, ~ 2.7 Ga (very thick haze)

upper-end haze, $\tau(550\text{nm}) \sim 1.5$: CH_4/CO_2 well above 0.2

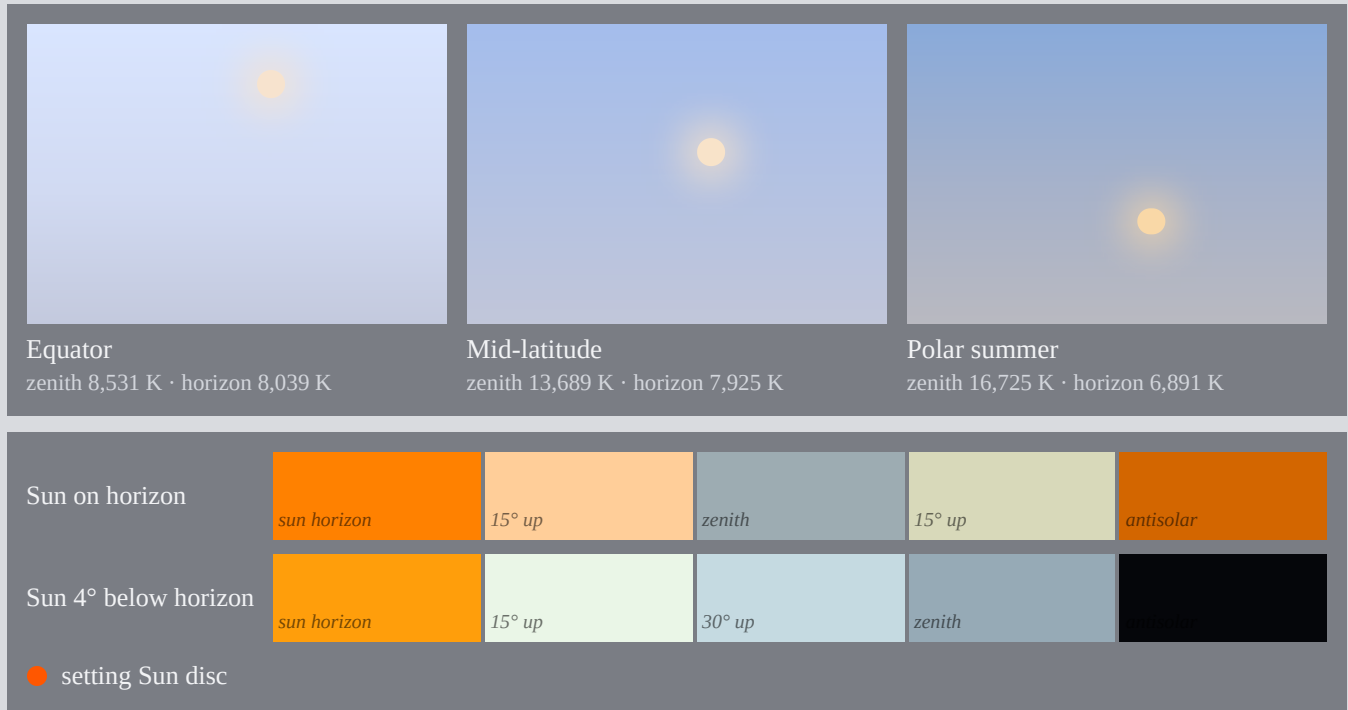


At the upper end of plausible haze ($\tau \sim 1.5$) the sky is a uniform saturated apricot ($3,300$ – $4,200$ K), fully Titan-like, and the polar Sun is reduced to an orange smear. Anything below this haze would have had roughly 20–30% of the surface sunlight of a clear sky.

2.2 Ga

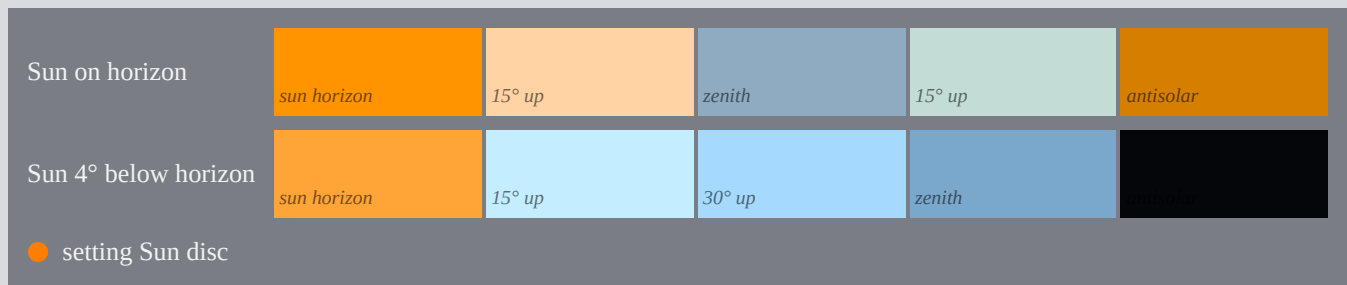
Paleoproterozoic, ~ 2.2 Ga

after the Great Oxidation Event: O₂ ~1% PAL, first ozone layer (~150 DU)



After the Great Oxidation Event a few percent of today's O₂ builds an ozone column of order 150 DU, about half the modern value. Noon skies are essentially modern blue. Twilight recovers its blue zenith, though a paler one than today's. (The pink antisolar arch, by contrast, needs no ozone: it is a Rayleigh effect and shows up faintly even in the Hadean model.) From here to the present, the daytime sky is blue everywhere; what changes is the aerosol load, the surface albedo, and the Sun's brightness.

700 Ma **Snowball Earth, ~700 Ma**
cold, dry, very clean air; ice/snow surface (albedo 0.75); some dust

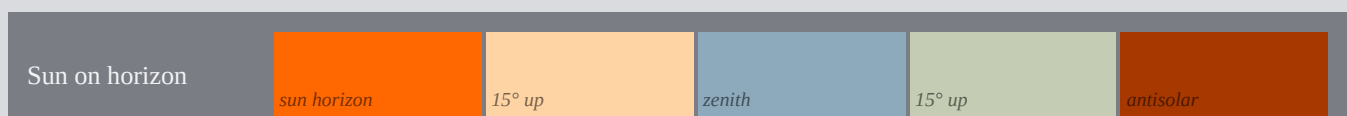


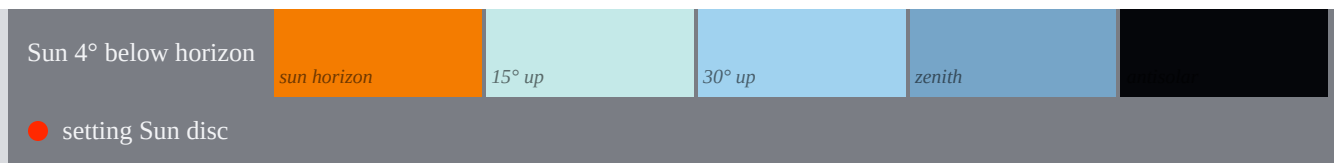
A frozen planet has a dry, exceptionally clean atmosphere and a surface albedo near 0.75. The result is the bluest sky in Earth's history: 9,000 K at the equatorial zenith and over 20,000 K over polar ice, with horizons that stay distinctly blue instead of whitening, because the sunlit ice floods the sky from below with blue-white light. Sunsets are golden and comparatively pale; there is little aerosol to redden them.

300 Ma

Carboniferous, ~300 Ma

O₂ ~30-35%; total pressure ~1.1 bar; wildfire smoke common





Thirty-plus percent O₂ raises the total pressure and Rayleigh scattering by ~10%, so skies are marginally brighter and horizons a touch whiter than today; the effect is subtle. Frequent wildfire smoke in the high-O₂ world would have produced red suns and brown-orange horizons far more often than in the modern era.

66 Ma

K-Pg impact winter, 66 Ma

months after Chicxulub: stratospheric soot (tau~1.5) + sulfate (tau~0.5)



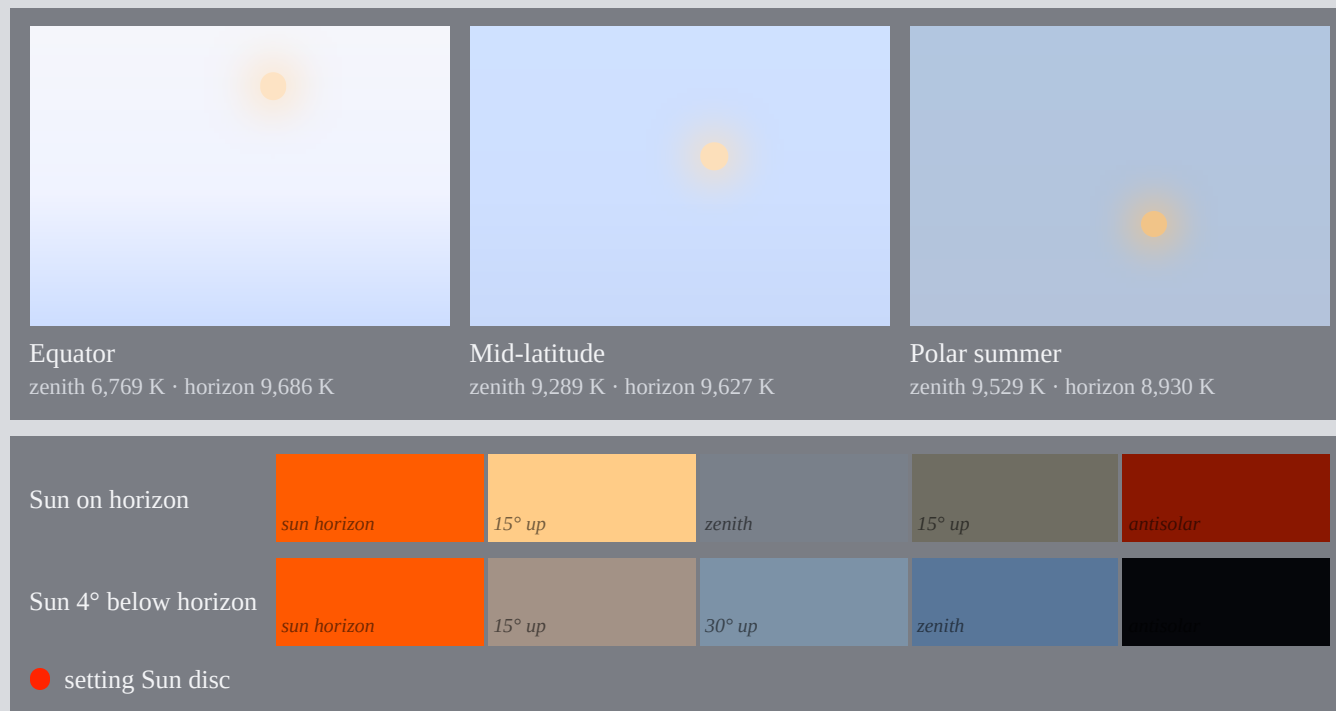
Months after the Chicxulub impact, a global stratospheric layer of soot and sulfate turns the sky into a dim, uniform amber-beige (4,200–5,100 K) with no blue at all, about one-fifth as bright as a clear sky at the equator and 20× dimmer at high latitudes. The Sun is a pale orange disc at low latitudes and

effectively gone near the poles. Sunsets do not happen as colour events: the whole dome simply fades to dark brown, and the twilight sky is grey.

1815

Volcanic year (Tambora 1815 / Toba-lite)

stratospheric sulfate veil, $\tau(550\text{nm}) \sim 0.4$

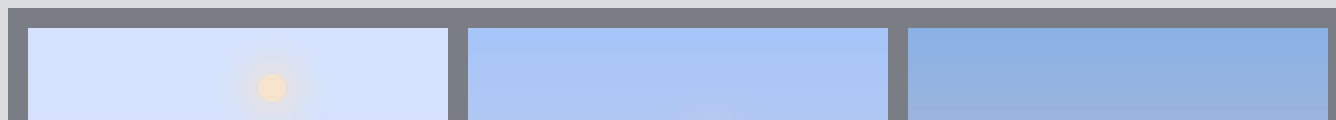


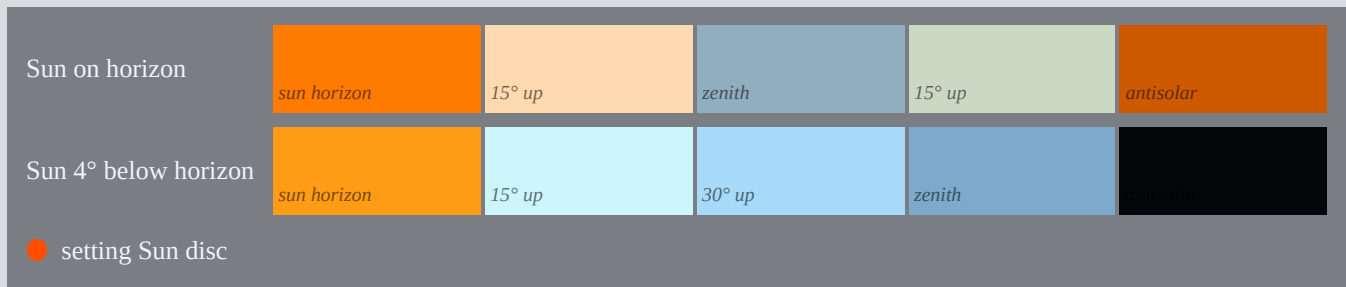
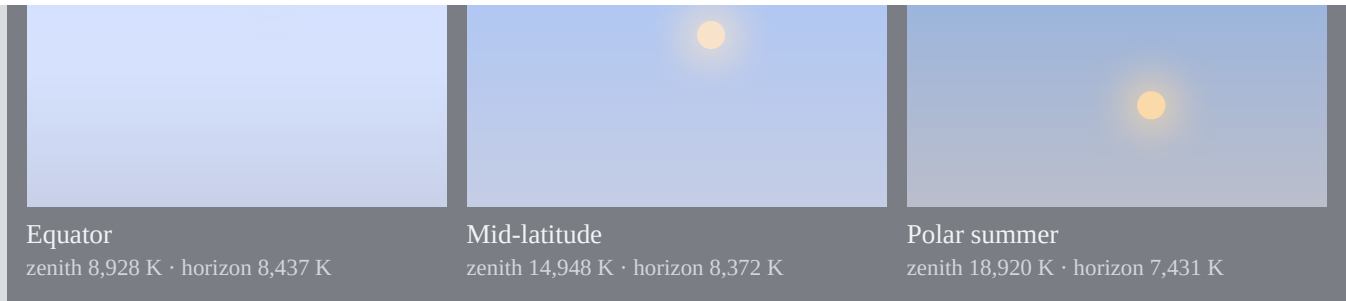
A Tambora-scale sulfate veil ($\tau \approx 0.4$) is what most people picture as a spectacular sky, and the model agrees. Noon skies are milky, whiter at the zenith than the horizon (the reverse of a clean sky). Sunsets glow salmon-orange, and the after-sunset sky 15–30° above the horizon turns a luminous pink-lavender: the high sulfate layer is still lit when the lower atmosphere is in shadow. This is the sky of Turner's 1816 paintings and of the Krakatoa twilights of 1883.

today

Modern, clean air

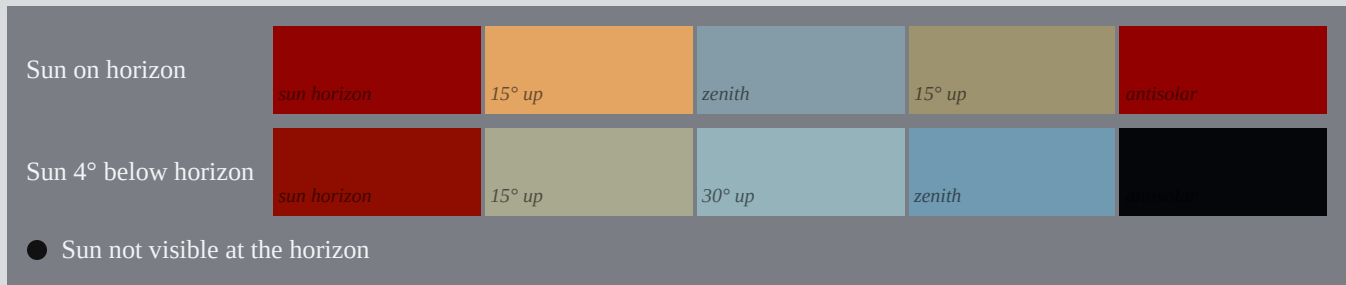
aerosol optical depth 0.1, ozone 300 DU





Baseline. Equatorial zenith ~9,000 K; mid-latitude ~15,000 K; polar summer ~19,000 K. Horizons run 7,400–8,400 K, essentially white. Setting Sun ~1,700 K; solar horizon orange; antisolar horizon pink.

today **Modern, polluted megacity**
aerosol optical depth 0.6, slightly absorbing



Aerosol optical depth 0.6 flattens the sky to a near-uniform pale grey-blue (6,500–8,300 K), brighter and whiter at the zenith than a clean sky, with the horizon slightly bluer than the zenith. The Sun disappears into grey murk well before it reaches the horizon.

Caveats

Paleoatmospheric compositions carry order-of-magnitude uncertainty, and the haze optical properties are taken from Titan-analog tholins; the three haze cases bracket that uncertainty. Aerosol loads for every pre-Cenozoic epoch are educated guesses. The colour-matching functions use an analytic fit, and colours are shown without chromatic adaptation, so the tints represent what a modern camera set to daylight balance would record rather than what an adapted observer would "see" (an adapted eye would perceive the hazy Archean sky as nearer to white and the modern sky as bluer than shown). Clouds are omitted throughout. The two-stream multiple-scattering term is approximate for the thickest atmospheres, where a full Monte Carlo treatment would refine the exact tint of the Hadean sky.

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